II: Contracted Farey expns

- Analogous to S-expns, we couple:
  (i) induced transformations of $(\Omega, \mathcal{B}, \alpha, \mathcal{F})$
  (ii) contraction (Seidel, 1855)

Contraction

- Given a GCF

(*) $$[d_0/e_0 ; e_1/d_1, e_2/d_2, \dots],$$

set

$$B_k := \begin{pmatrix} 0 & e_k \\ 1 & d_k \end{pmatrix}, \quad k \geq -1,$$

where $e_{-1} = 1$, $d_{-1} = 0$, and

$$\begin{pmatrix} P_{[j,k-1]} & P_{[j,k]} \\ Q_{[j,k-1]} & Q_{[j,k]} \end{pmatrix} = B_{[j,k]} := B_j \cdots B_k, \quad -1 \leq j \leq k.$$

For $j = -1$, use special notation

$$\begin{pmatrix} P_{k-1} & P_k \\ Q_{k-1} & Q_k \end{pmatrix} := B_{[-1,k]}.$$

- Note: $\dfrac{P_k}{Q_k} = B_{[-1,k]} \cdot 0 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}\begin{pmatrix} 0 & e_0 \\ 1 & d_0 \end{pmatrix} \cdots \begin{pmatrix} 0 & e_k \\ 1 & d_k \end{pmatrix} \cdot 0$

$$= \cfrac{1}{0 + \cfrac{e_0}{d_0 + \ddots \cfrac{e_k}{d_k}}} = [d_0/e_0 ; e_1/d_1, \dots, e_k/d_k],$$

and

$$(**)\qquad \begin{pmatrix} P_{k+1} \\ Q_{k+1} \end{pmatrix} = B_{[-1,k]} \begin{pmatrix} e_{k+1} \\ d_{k+1} \end{pmatrix} = \begin{pmatrix} d_{k+1} P_k + e_{k+1} P_{k-1} \\ d_{k+1} Q_k + e_{k+1} Q_{k-1} \end{pmatrix}.$$

- Defn.: Suppose the convergents P_n/Q_n of (*) are pairwise distinct. The <u>contraction</u> of (*) w/r/t a strictly increasing sequence $(n_k)_{k\geq 0}$ in $\mathbb{N}$ is the GCF $[d_0'/e_0'; e_1'/d_1', e_2'/d_2', \dots]$ with

$$\begin{pmatrix} e_{k+1}' \\ d_{k+1}' \end{pmatrix} := \begin{pmatrix} -\det\left(B_{[n_{k-1}+2,\, n_{k+1}]}\right) Q_{[n_{k-2}+2,\, n_{k-1}]} \, Q_{[n_{k-1}+2,\, n_k]} \\ Q_{[n_{k-1}+2,\, n_{k+1}]} \end{pmatrix},$$

$k \geq -1$, where $n_k = k$ for $k < 0$.

- Thm. (Seidel, 1855): The contraction of (*) w/r/t $(n_k)_{k\geq 0}$ has convergents P_{n_k}/Q_{n_k}.

- Idea: Induction using (**) and

$$Q_{[j+1,k]} = \frac{P_{j-1} Q_k - Q_{j-1} P_k}{\det B_{[-1,j]}}.$$

□

(- Remarks:
  - contraction is <u>much</u> broader than singularisation
  - $P_k' = c_k P_{n_k}$, $Q_k' = c_k Q_{n_k}$, where $c_k = \prod_{j=0}^{k-1} Q_{[n_j+2,\, n_j]}$)

Inducing $\mathcal{F}$

- Throughout, $R \subset \Omega$ is measurable, $\bar{u}(\mathring{R}) > 0$, and $\bar{u}(R) < \infty$ or $R = \Omega$.

- Defn.: The <u>hitting time</u> $N_R : \Omega \to \mathbb{N} \cup \{\infty\}$ is

$$N_R(z) := \inf\{ n \geq 1 \mid \mathcal{F}^n(z) \in R \}$$

(Remark: $\mathcal{F}$ is conservative $[\bar{u}(A) > 0 \Rightarrow \bar{u}(A \cap (\bigcup_{n \geq 0} T^{-n} A)) > 0]$ and ergodic $\Rightarrow \mathcal{F}^n(z) \in R$ i.o. $\bar{u}$-a.s.)

Let $\mathcal{F}_R := \mathcal{F}^{N_R} : \Omega \to R$. The <u>induced transformation</u> is $(R, \mathcal{B}_R, \bar{u}_R, \mathcal{F}_R)$, where

$$\bar{u}_R(S) := \bar{u}(S) / \bar{u}(R), \quad S \subset R.$$

- Fact! $\bar{u}_R$ is an ergodic a.c.i.p. for $\mathcal{F}_R$

- Prop.: The (metric) entropy is $h_{\bar{u}_R}(\mathcal{F}_R) = \dfrac{\pi^2}{6\,\bar{u}(R)}$.

- Recall: $\mathcal{F}(x, y) = \left(A_{\varepsilon(x)}^{-1} \cdot x,\ A_1 A_{\varepsilon(x)}^{T} A_1^{-1} \cdot y \right)$, so

$$\mathcal{F}^n(x, y) = \Big(\underbrace{A_{\varepsilon_n}^{-1} \cdots A_{\varepsilon_1}^{-1}}_{A_{[0,n]}^{-1} = \begin{pmatrix} u_n & t_n \\ s_n & r_n \end{pmatrix}^{-1}} \cdot x,\ A_1 \underbrace{A_{\varepsilon_n}^{T} \cdots A_{\varepsilon_1}^{T}}_{A_{[0,n]}^{T}} A_1^{-1} \cdot y \Big).$$

- Setting $A_R(z) = \begin{pmatrix} u_R(z) & t_R(z) \\ s_R(z) & r_R(z) \end{pmatrix} := A_{[0, N_R(z)]}$ gives

$$\mathcal{F}_R(z) = \left(A_R^{-1}(z) \cdot x,\ A_1 A_R^T(z) A_1^{-1} \cdot y \right).$$

- Set $\dfrac{u_n^R}{s_n^R} = \dfrac{u_{N_n^R}}{s_{N_n^R}}$ where $N_n^R = \sum_{k=0}^{n-1} N_R(\mathcal{F}_R^k(z))$

satisfies $\mathcal{F}_R^n(z) = \mathcal{F}^{N_n^R}(z)$.

Contracted Farey expn's

- Recall the Farey expn.

$$x = \left[(1-\varepsilon_1)/1;\ (-1)^{\varepsilon_1+1}/2\varepsilon_2,\ (-1)^{\varepsilon_2+1}/2\varepsilon_3, \dots \right],$$

whose n^{th} convergent is u_{n+1}/s_{n+1}.

- Defn: The <u>contracted Farey expn</u> (CFE) of x w/r/t R, denoted

$$\left[d_0^R/e_0^R;\ e_1^R/d_1^R,\ e_2^R/d_2^R, \dots \right],$$

is the contraction of the Farey expn. of x w/r/t $(N_{k+1}^R - 1)_{k \geq 0}$, where $N_k^R = N_k^R(x,1)$.

- Cor.: The kth convergent of the CFE is U_{k+1}^R / S_{k+1}^R.

  - Proof: By defn. and Seidel's Thm., the kth convergent is $U_{N_{k+1}^R}^R / S_{N_{k+1}^R}^R = U_{k+1}^R / S_{k+1}^R$. □

- Digits Prop.: Set $z_k^R := \mathcal{T}_R^k(x,1)$. Then

$$\begin{pmatrix} e_0^R \\ d_0^R \end{pmatrix} = \begin{pmatrix} S_R(z_0^R) \\ U_R(z_0^R) \end{pmatrix}, \quad \begin{pmatrix} e_{k+1}^R \\ d_{k+1}^R \end{pmatrix} = \begin{pmatrix} -\det\left(A_R(z_k^R)\right) S_R(z_{k-1}^R)\, S_R(z_{k+1}^R) \\ S_R(z_k^R)\, U_R(z_{k+1}^R) + r_R(z_k^R)\, S_R(z_{k+1}^R) \end{pmatrix}.$$

- Idea: Compare entries of

$$B_{[j,k]} = \begin{pmatrix} 0 & (-1)^{\varepsilon_{j+1}} \\ 1 & 2-\varepsilon_{j+1} \end{pmatrix} \cdots \begin{pmatrix} 0 & (-1)^{\varepsilon_{k+1}} \\ 1 & 2-\varepsilon_{k+1} \end{pmatrix}$$

with

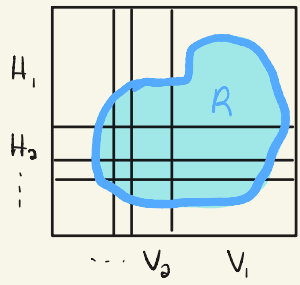
$$A^R_{[0,k-j]}(z_j^R) = \begin{pmatrix} u^R_{kj}(z_j) & t^R_{kj}(z_j) \\ s^R_{kj}(z_j) & r^R_{kj}(z_j) \end{pmatrix} := A_{\varepsilon_{N_{j+1}^R}} \cdots A_{\varepsilon_{N_k^R}}.$$

In particular, show

$$\det B_{[N_k^R+1,\, N_{k+1}^R-1]} = \det A^R_{[0,1]}(z_k^R) = \det A_R(z_k^R),$$

$$Q_{[N_j^R+1,\, N_k^R-1]} = S^R_{k-j}(z_j^R), \quad 0 \le j < k. \qquad \square$$

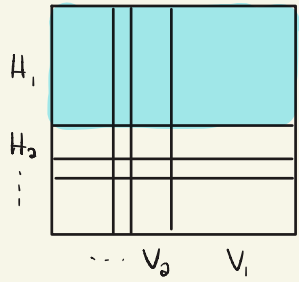
- Remark: By Digits Prop., each $(R, \mathcal{B}, \bar{\mu}_R, \mathcal{T}_R)$ determines a CF-algorithm producing a CFE of (a.e.) x



R CFs

- Thm. (Brown – Yin, 1996): With $R = H_1$,

$$(R, \mathcal{B}_R, \bar{\mu}_R, \mathcal{T}_R) \cong (\Omega, \mathcal{B}_\Omega, \bar{\mu}_G, \mathcal{G})$$



- Sketch: Let $z = (x,y) \in R$, $a(x) = \lfloor 1/x \rfloor$. Then

$$A_R(z) = A_0^{a(x)-1} A_1 = \begin{pmatrix} 0 & 1 \\ 1 & a(x) \end{pmatrix}.$$

Letting $\phi : R \to \Omega$,
$(x,y) \mapsto (x, A_1^{-1} \cdot y)$,

$$\begin{array}{ccc} R & \xrightarrow{\mathcal{T}_R} & R \\ \phi \downarrow & & \downarrow \phi \\ \Omega & \xrightarrow[\mathcal{G}]{} & \Omega \end{array}$$

$$\begin{aligned} \phi^{-1} \circ \mathcal{G} \circ \phi (x,y) &= \phi^{-1} \circ \mathcal{G}(x, A_1^{-1} \cdot y) \\ &= \phi^{-1}\left(\begin{pmatrix} 0 & 1 \\ 1 & a(x) \end{pmatrix}^{-1} \cdot x, \begin{pmatrix} 0 & 1 \\ 1 & a(x) \end{pmatrix}^{T} \cdot A_1^{-1} \cdot y \right) \\ &= \mathcal{T}_R(x,y). \end{aligned}$$

One also finds $\bar{\mu}_G = \phi_* \bar{\mu}_R$. □

- Suppose $x = [0; a_1, a_2, \ldots]$, and let

$$z_k^R = (x_k^R, y_k^R) = \mathcal{F}_R^k(x, 1)$$

Then

$$A_R(z_k^R) = \begin{pmatrix} 0 & 1 \\ 1 & a(x_k^R) \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ 1 & a_{k+1} \end{pmatrix},$$

and by Digits Prop.,

$$\begin{pmatrix} e_0^R \\ d_0^R \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad \begin{pmatrix} e_{k+1}^R \\ d_{k+1}^R \end{pmatrix} = \begin{pmatrix} -(-1)\cdot 1 \cdot 1 \\ 1\cdot 0 + a_{k+1}\cdot 1 \end{pmatrix} = \begin{pmatrix} 1 \\ a_{k+1} \end{pmatrix}.$$

Hence the CFE

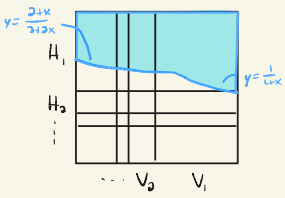
$$x = [0/1;\ 1/a_1,\ 1/a_2, \ldots]$$

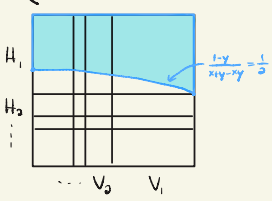
recovers the RCF-expn.

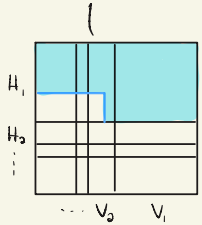
S-expns

- S-expns couple
  (i) induced transformations of $\mathcal{G} \cong \mathcal{F}_{H_1}$,
  (ii) ~~singularisation~~ contraction of RCFs.

- Hence S-expns are CFEs. These include optimal, diagonal, α-CFs, $\alpha \geq 1/2$.







α-CFs

- Recall $([\alpha-1,\alpha), \mathcal{B}, \nu_\alpha, G_\alpha)$, where for $u \in [\alpha-1,\alpha)\setminus\{0\}$,

$$G_\alpha(u) = \frac{1}{|u|} - \left\lfloor \frac{1}{|u|} \right\rfloor_\alpha, \qquad \lfloor y \rfloor_\alpha = \lfloor y + 1 - \alpha \rfloor.$$

- Goal: Sketch of the following:

- Thm. 1 (Dajani-Kraaikamp-S. 2026): For each $\alpha \in (0, 1]$, $\exists$ explicit $R \subset \Omega$ s.t. $(R, \mathcal{B}_R, \bar{\nu}_R, \mathcal{F}_R)$ is a natural extn. (NE) of $([\alpha-1,\alpha), \mathcal{B}, \nu_\alpha, G_\alpha)$.

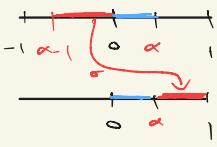
(- Case $\alpha = 0$ can be treated seperately using Adler-Flatto (1983) and Heersink (2019): G_0 realised as induced transformation of double cover of $\mathcal{F}$.)

- Suffices to consider $\alpha \in (0, \frac{1}{2})$. Define

$$H_\alpha : [0,1) \to [0,1)$$
$$x \mapsto \sigma \circ G_\alpha \circ \sigma^{-1}(x),$$

where

$$\sigma : [\alpha-1, \alpha) \to [0,1)$$
$$u \mapsto u \pmod 1$$



- Lem. 1: For $x = [0; a_1, a_2, \dots]$,

$$H_\alpha(x) = \begin{cases} F^{a_1}(x) = G_1(x), & x < \alpha \\ F(x), & \alpha \le x \le 1/2 \\ F^{a_1+a_2}(x) = G_1^2(x), & 1/2 < x \end{cases}$$

- Proof: https://www.desmos.com/calculator/tj0lrny49b □

- Idea: Construct $R \subset \Omega$ s.t.

$$\hat{F}_R(x,y) = \begin{cases} \hat{F}^{a_1}(x,y), & x < \alpha, \\ \hat{F}(x,y), & \alpha \le x \le 1/2, \\ \hat{F}^{a_1+a_2}(x,y), & 1/2 < x. \end{cases}$$

- Then, if $\pi(x,y) = x$,

$\pi \circ \hat{F} = F \circ \pi$ and Lem. 1 $\Rightarrow$

$\pi \circ \hat{F}_R = H_\alpha \circ \pi = \sigma \circ G_\alpha \circ \sigma^{-1} \circ \pi$.

Setting $\phi := \sigma^{-1} \circ \pi : R \to [\alpha-1, \alpha)$,

$\phi \circ \hat{F}_R = G_\alpha \circ \phi$.

$$\begin{array}{ccc} R & \xrightarrow{\hat{F}_R} & R \\ {\scriptstyle\phi}\downarrow & & \downarrow{\scriptstyle\phi} \\ [\alpha-1,\alpha) & \xrightarrow{G_\alpha} & [\alpha-1,\alpha) \end{array}$$

Further, ϕ is

- surjective (clear after R constructed)
- measurable (σ^{-1}, π measurable)

· measure-preserving ($\phi_* \bar{\mu}_R$ an a.c.i.p., so $\phi_* \bar{\mu}_R = \nu_\alpha$ by uniqueness; hence $\nu_\alpha(A) = \phi_* \bar{\mu}_R(A) = \bar{\mu}_R(\phi^{-1}(A))$.)

and

· $\mathcal{B}_R = \bigvee_{k=0}^{\infty} \mathcal{F}_R^k(\phi^{-1} \mathcal{B}_{\Sigma(\alpha-1,\alpha)})$

(shown by considering 'cylinder sets').

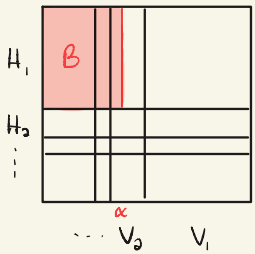
· R constructed in 3 steps:

1) Define $B := [0, \alpha) \times [1/2, 1]$, $k : H_1 \to \mathbb{N}$ by

$$k(z) := \inf\{j > 0 \mid \mathcal{F}_{H_1}^j(z) \in B\}$$

· Exercise 1:

$$k(\mathcal{F}_{H_1}(z)) = \begin{cases} 1, & z \in B \\ k(z)+1, & z \notin B \end{cases}$$

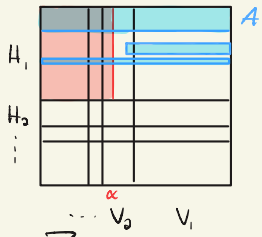


2) Define $A := \{z \in H_1 \mid k(z) \text{ is odd}\}$.

· Exercise 2: The 1st return $N_A : A \to \mathbb{N}$ of $\mathcal{F}$ to A is

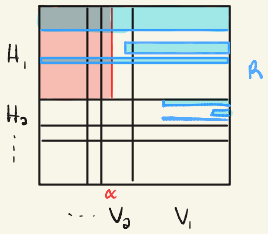
$$N_A(z) = \begin{cases} a_1, & z \in A \cap B \\ a_1 + a_2, & z \in A \setminus B \end{cases},$$

where $z = (x, y)$, $x = [0; a_1, a_2, \ldots]$.



3) Set $A_a := (A \setminus B) \cap V_a$, $a \geq 1$, and

$$R := A \sqcup \bigsqcup_{a>1} \bigsqcup_{\lambda=1}^{a-1} \mathcal{T}^{\lambda}(A_a).$$



- Exercise 3: The 1st return $N_R : R \to \mathbb{N}$ of $\mathcal{T}$ to R is

$$N_R(z) = \begin{cases} a_1, & x < \alpha, \\ 1, & \alpha \leq x < 1/2, \\ a_1 + a_2, & 1/2 < x. \end{cases}$$

- Since $\mathcal{T}_R = \mathcal{T}^{N_R}$, we've constructed R as desired.

(- Remark: If $z = (x, y) \in H_1$ with $y = [0; 1, b_1, b_2, \ldots]$, $\frac{1}{b_1} < \alpha$, then $\mathcal{T}_{H_1}^{-1}(z) \in B \Rightarrow z \in A$. Hence $[0,1) \times \{y\} \subset A \Rightarrow \phi = \sigma^{-1} \circ \sigma$ surjective.)

(- Remark: Using Digits Prop., one can show the CFE of x w/r/t R coincides with the α-CF-expn. of x.)